Sewon Kim

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wontothree.github.io

Research Interests

Optimal Control and **Robot Learning**. I am interested in developing scalable learning and control methods and their theoretical foundations for robotics.

Education

University of Seoul Mar 2022 - Aug 2027
B.S. in Electrical and Computer Engineering, *Minor* in Mathematics Seoul, South Korea

Experience

Robotics and Physical AI Lab, Yonsei University Aug 2026 - Present
Intern (Advisor: Yonghyeon Lee); Dexterous Manipulation Seoul, South Korea

PDE and AI Lab, Yonsei University Sep 2024 - Present
Intern (Advisor: Dohyun Kwon); Trilevel Optimization Seoul, South Korea

InBody AI Scale Team, InBody Jan 2024 - Jan 2025
Computer Vision Engineer; InBody AI Scale [P4][C1] Seoul, South Korea

Control and Dynamic System Lab, University of Seoul Mar 2023 - Dec 2024
Intern (Advisor: Gyunghoon Park); F1TENTH [P2] Seoul, South Korea

Projects

[P1] **NAMUH X A1: Autonomous Air Purification Agent** Aug 2025 - Sep 2025
4th Place (/200) ($2,200) — SK NAMUHX A1: Autonomous Navigation and Path Optimization; Independently design and implemented full-stack autonomous air purification agent in NumPy from scratch, covering Monte Carlo localization, MPPI, and traveling salesman problem.

[P2] **F1TENTH: Autonomous Racing with MPPI** Mar 2024 - Dec 2024
5th Place (/12) — 22nd F1tenth Autonomous Grand Prix at CDC Milan 2024; Built full-stack physical autonomous navigation system for racing, covering SLAM, MCL, Raceline Opt, MPPI. Led the project with 2 undergrads.

[P3] **Cart Pole System Design and Control with NMPC** Jul 2024 – Sep 2024
3rd Place (/25) ($220) — UOS ECE Innovation Fair; Independently developed a physical Cart-Pole system with $272, covering modeling, motor control, sensor integration, state estimation, embedded systems, and MPC.

[P4] **InBody AI Scale** Jan 2024 - Jan 2025
Consumer Electronic Show 2025; Developed height estimation model (acc 99.4%) and face recognition system of "InBody AI Scale", automated body composition analyzer that reduces measurement time from 120s to 40s.

[P5] **ECG data-driven cardiovascular diagnosis device** Jul 2023 - Jul 2023
3rd Place (/12) ($1450) — InBody Hackathon 2023; Developed a fast prototype.

[P6] **Autonomous mission mobile robot** Jul 2023 - Jul 2023
3rd Place (/20) ($220) — Object Detection-Based Solar Tracking Robot Challenge, Korea University

[P7] **Sidaeting** Mar 2022 - Jun 2022
1,700 User; Implemented a matching algorithm that combinatorially optimizes collective happiness from

users' ideal type data and developed the frontend for "Sidaeting", a dating service launched by UOSLIFE.

Patents

[C1] Kichul Cha, Jaeho Kim, Seungseo Park, Suhyun Ga, Wonjun Lee, and Sewon Kim, "Electronic Device for Determining Body Composition of User and Method for Operating the Same," KR Patent Application No. 10-2024-0125030 (Publication No. 10-2026-0038628), filed Sep. 12, 2024.

Skills

- Languages: Korean, English
- Programming Languages: C, Cpp, Java, JavaScript, Julia, Python, SQL, HTML, CSS
- Frameworks: Isaac Sim/Lab, ROS 1/2 (Melodic, Noetic, Foxy, Humble, Jazzy), MuJoCo, PyTorch

Services

Institute of Innovation for Future Army, Republic of Korea Army — Jan 2025 - Jul 2026
Drone Researcher; Autonomous Drone for Indoor Mapping in CQB — Daejeon, South Korea

AI Robotics KR — Sep 2023 - Sep 2024
Management Team; hosted "Robotalk", "Human and AI Robotics Together" — Seoul, South Korea

Let's Play Software, NAVER Connect Foundation — Apr 2022 - Jan 2023
Elementary School Software Instructor — Seoul, South Korea